

HPC School



HPC Primer - SSH

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SSH

The Slides in PDF:

<https://edu.nl/98drq>

whoami Instructors

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Requirements I

A SURF Cua account

<https://portal.cua.surf.nl>

A terminal on Snellius

<https://ondemand.snellius.surf.nl>

Basic terminal (bash) knowledge

SSH : Secure Shell

A protocol for securely accessing and managing remote systems over a network.

For using HPC/Supercomputers via the command line (CLI), for example from your laptop

Access to online Cloud services like github.com.

Why SSH

- Verify the Identity of remote systems
- Verify the Identity of the user by the remote system
- Encrypt network traffic
- Password less login

Asymmetric Encryption

public key

- Used for encryption that only private key can decrypt
- Can be given (published) to everyone.

private key

- Remains (only) with the user
 - The possession of this key is proof of the user's identity
- Used to generate public key.
- Used to decrypt what was encrypted with the public key
- Password Protected! (not SHOULD, MUST)

Asymmetric Encryption



Generating a key pair (CLI)

ssh-keygen : generate unique Pub/Priv Key

```
$ ssh-keygen -t ed25519 -C "<EMAIL>"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/Users/alain/.ssh/id_ed25519): <ENTER>
Enter passphrase for "/Users/alain/.ssh/id_ed25519" (empty for no passphrase): <PASSWORD>

$ ls -l $HOME/.ssh/

-rw----- 1 alain  staff  1766 Sep 28  2026 id_ed25519
-rw----- 1 alain  staff   394 Sep 28  2026 id_ed25519.pub
```

Generating a key pair walk trough

<https://supercomputing.tue.nl/documentation/steps/access/ssh/>

(<https://edu.nl/p4vtc>)

Using your key (CLI)

`ssh-copy-id` (Linux/Mac): place public key on the correct place on remote system

On remote system key is automatically added to:

```
$HOME/.ssh/authorized_keys
```

Examples (Linux)

```
ssh-copy-id <username>@snellius.surf.nl  
ssh-copy-id <username>@hpc.tue.nl
```

Then

```
ssh <username>@snellius.surf.nl  
ssh <username>@hpc.tue.nl
```

No longer a password asked (but maybe for your private key)

Loading your private key into memory

```
ssh-agent bash  
ssh-add
```

The password for your private keys is asked,
but after that the loaded key is used,
as long as your session lasts

scp : copy a file to a remote system using SSH

```
scp .ssh/id_ed25519.pub <name>@hpc.tue.nl:~/.ssh/authorized_keys
```

SSH "one-shot" commands

```
# Check the files in your home directory on snellius
```

```
$ ssh tesr52960@snellius.surf.nl ls -l
```

```
total 70
```

drwxr-x---	2	tesr52960	tesr52960	4096	Oct	8	2025	bin
drwxr-x---	4	tesr52960	tesr52960	4096	Oct	10	2024	snellius-hybrid-jobs
drwxr-xr-x	2	tesr52960	tesr52960	4096	Jan	15	2025	Desktop
drwxr-xr-x	3	tesr52960	tesr52960	4096	Mar	9	2026	Documents
drwxr-xr-x	2	tesr52960	tesr52960	4096	Jan	15	2025	Downloads
-rw-r-----	1	tesr52960	tesr52960	1550	Apr	10	16:01	fluent-950150.trn
-rw-r-----	1	tesr52960	tesr52960	6028	Apr	13	13:50	fluent103193.trn
drwxr-x---	2	tesr52960	tesr52960	4096	Nov	11	2025	GIT

Creating \$HOME/.ssh/authorized_keys

```
ssh curxxxx@snellius.surf.nl # Password curxxxx?
```

```
cd .ssh  
cp id_ed25519.pub authorized_keys
```

```
exit  
ssh curxxxx@snellius.surf.nl # Password private key?
```

We still need a password!

```
exit  
ssh-agent bash  
ssh-add
```

And now we don't:

```
ssh curxxxx@snellius.surf.nl  
exit
```

```
cat id_ed25519.pub  
cd ..
```

SSH Assignment I

Hands on

1. Goto. <https://ondemand.snellius.surf.nl>
2. Open a terminal on Snellius

SURF

Apps ▾

Files ▾

Jobs ▾

Clusters ▾

Interactive Apps ▾

SURF

 Module Browser

 Snellius Shell Access

 System Status

OnDemand provides an integrated, single access point

Pinned Apps A featured subset of [all](#)

Interactive Apps

3. Create a ssh key public/private keypair
4. Copy the Public key to test@hpcschool.nl (T3stT3st%)
5. Check what files are there

SSH Assignment Solution

```
$ ssh-keygen -t ed25519 -C "Test Key"  
$ ssh-copy-id test@hpc-school.nl  
$ bash ssh-add  
$ ssh test@hpc-school.nl ls -l
```

SSH Assignment II

1. Copy Job1.sbatch from hpcschool.nl
2. Copy Job1.sbatch to your local machine from Snellius

SSH Assignment Solution

```
scp test@hpc-school.nl:~/Job1.sbatch .  
Download via ondemand.snellius.surf.nl (files menu)
```